STAT 201A Homework 1 Solutions

Theorem numbers refer to Casella & Berger (C&B). The two results used most often are

- **Theorem 1.2.9.** If P is a probability function and A and B are any sets in $\mathcal{B}$, then
  - **a.** $P(B \cap A^c) = P(B) - P(A \cap B)$;
  - **b.** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$;
  - **c.** If $A \subset B$, then $P(A) \leq P(B)$.
- **Double containment:** to show $E = F$, show $E \subset F$ and $F \subset E$, i.e., $x \in E \Rightarrow x \in F$ and $x \in F \Rightarrow x \in E$.

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1. (Exercise 1.10) *Formulate and prove a version of DeMorgan's Laws that applies to a finite collection of sets $A_1, \ldots, A_n$.*

Statement. For any sets $A_1, \ldots, A_n$,

$$\text{(a)} \left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c, \qquad \text{(b)} \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c.$$

Proof of (a). Each step below is an "if and only if," so it proves both containments at once:

$$\begin{aligned} x \in \left(\textstyle\bigcup_{i=1}^{n} A_i\right)^c &\iff x \notin \textstyle\bigcup_{i=1}^{n} A_i \\ &\iff \text{it is not true that } x \in A_i \text{ for some } i \\ &\iff x \notin A_i \text{ for every } i = 1, \ldots, n \\ &\iff x \in A_i^c \text{ for every } i \\ &\iff x \in \textstyle\bigcap_{i=1}^{n} A_i^c. \end{aligned}$$

Proof of (b). Apply (a) to the sets $A_1^c, \ldots, A_n^c$ and use $(A^c)^c = A$:

$$\left(\bigcup_{i=1}^{n} A_i^c\right)^c = \bigcap_{i=1}^{n} A_i.$$

Taking complements of both sides gives $\bigcup_{i=1}^{n} A_i^c = \left(\bigcap_{i=1}^{n} A_i\right)^c$. □

Alternative: induction on n, using the two-set laws (Theorem 1.1.4(d)) for the base case and the step $\bigcup_{i=1}^{n+1} A_i = \left(\bigcup_{i=1}^{n} A_i\right) \cup A_{n+1}$.

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2. (Exercise 1.11(c), extended)

(a) Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be sigma algebras on S, and let $\mathcal{B} = \mathcal{B}_1 \cap \mathcal{B}_2$, the collection of sets that belong to **both**. We check the three properties of Definition 1.2.1.

1. $\emptyset \in \mathcal{B}_1$ and $\emptyset \in \mathcal{B}_2$, so $\emptyset \in \mathcal{B}$.
2. If $A \in \mathcal{B}$, then $A \in \mathcal{B}_1$, so $A^c \in \mathcal{B}_1$; likewise $A^c \in \mathcal{B}_2$. Hence $A^c \in \mathcal{B}$.
3. If $A_1, A_2, \cdots \in \mathcal{B}$, then all $A_i \in \mathcal{B}_1$, so $\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}_1$; likewise for $\mathcal{B}_2$. Hence $\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$.

So $\mathcal{B}_1 \cap \mathcal{B}_2$ is a sigma algebra. □ (The same argument works for the intersection of any collection of sigma algebras.)

(b) No. Let $S = \{1, 2, 3\}$ and

$$\mathcal{B}_1 = \{\emptyset, \{1\}, \{2, 3\}, S\}, \qquad \mathcal{B}_2 = \{\emptyset, \{2\}, \{1, 3\}, S\}.$$

Each is a sigma algebra (it is closed under complements and unions). But $\mathcal{B}_1 \cup \mathcal{B}_2$ contains $\{1\}$ and $\{2\}$ and not their union $\{1, 2\}$, so it is not closed under unions and is not a sigma algebra.

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3. (Exercise 1.13) *If $P(A) = \frac{1}{3}$ and $P(B^c) = \frac{1}{4}$, can A and B be disjoint?*

No. By Theorem 1.2.8(c), $P(B) = 1 - P(B^c) = \frac{3}{4}$. If A and B were disjoint, then by Axiom 3

$$P(A \cup B) = P(A) + P(B) = \frac{1}{3} + \frac{3}{4} = \frac{13}{12} > 1,$$

which contradicts $P(A \cup B) \leq 1$ (Theorem 1.2.8(b)). Indeed, Theorem 1.2.9(b) gives $P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq \frac{13}{12} - 1 = \frac{1}{12} > 0$.

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4. (Exercise 1.14) *If S has n elements, the number of subsets of S is 2^n.*

Write $S = \{s_1, \ldots, s_n\}$. A subset $A \subset S$ is determined by answering, for each $i = 1, \ldots, n$, the yes/no question "is $s_i \in A$?" Different subsets give different answer sequences, and every sequence of n answers gives a subset. So subsets correspond one-to-one to sequences in $\{\text{yes}, \text{no}\}^n$. By the Fundamental Theorem of Counting (Theorem 1.2.14), there are $2 \times 2 \times \cdots \times 2 = 2^n$ such sequences. □

Alternative: there are $\binom{n}{k}$ subsets of size k, so the total is $\sum_{k=0}^{n} \binom{n}{k} = (1+1)^n = 2^n$ by the binomial theorem. *Or* induction: adding an element s_{n+1} doubles the count, since every subset of $\{s_1, \ldots, s_{n+1}\}$ either omits s_{n+1} or is a subset of $\{s_1, \ldots, s_n\}$ with s_{n+1} added.

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5. *Prove that $P(A \cap B) \leq \min\{P(A), P(B)\} \leq \max\{P(A), P(B)\} \leq P(A \cup B)$.*

- **First inequality.** $A \cap B \subset A$ and $A \cap B \subset B$, so by Theorem 1.2.9(c), $P(A \cap B) \leq P(A)$ and $P(A \cap B) \leq P(B)$. A number below both $P(A)$ and $P(B)$ is below the smaller of the two.
- **Middle inequality.** The minimum of two numbers is at most their maximum.
- **Last inequality.** $A \subset A \cup B$ and $B \subset A \cup B$, so $P(A) \leq P(A \cup B)$ and $P(B) \leq P(A \cup B)$ by Theorem 1.2.9(c). A number above both is above the larger. □

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6. *Show that* $P(A \triangle B) = P(A) + P(B) - 2P(A \cap B)$.

The sets $A \cap B^c$ and $A^c \cap B$ are disjoint: a point in both would be in $B^c \cap B = \emptyset$. So by Axiom 3,

$$P(A \triangle B) = P(A \cap B^c) + P(A^c \cap B).$$

By Theorem 1.2.9(a), $P(A \cap B^c) = P(A) - P(A \cap B)$ and $P(B \cap A^c) = P(B) - P(A \cap B)$. Adding,

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B). \quad \square$$

Check: $P(A \triangle B) = P(A \cup B) - P(A \cap B)$, which is Theorem 1.2.9(b) minus $P(A \cap B)$. Compared with $P(A \cup B)$, the overlap is removed twice instead of once.

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7. *Let* $S = \{1, 2, 3, 4\}$. *List the smallest sigma algebra that contains* $\{1\}$ *and* $\{2\}$.

Any sigma algebra containing $\{1\}$ and $\{2\}$ must contain $\{1, 2\}$ (unions) and therefore $\{1, 2\}^c = \{3, 4\}$ (complements). The three sets $\{1\}$, $\{2\}$, $\{3, 4\}$ partition S, and the sigma algebra must contain all $2^3 = 8$ unions of them:

$$\mathcal{B} = \Big\{\emptyset,\ \{1\},\ \{2\},\ \{3,4\},\ \{1,2\},\ \{1,3,4\},\ \{2,3,4\},\ S\Big\}.$$

This collection is closed under complements (e.g., $\{1\}^c = \{2, 3, 4\}$) and under unions (a union of unions of the three blocks is again such a union), so it is a sigma algebra. It is the smallest because every one of its sets was forced. Note that $\{3\} \notin \mathcal{B}$: this sigma algebra cannot tell 3 from 4.

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8. Infinite unions and intersections

(a)

1. $\displaystyle\bigcup_{n=1}^{\infty} \left[\frac{1}{n}, 1 - \frac{1}{n}\right] = (0, 1)$. (The interval is empty for $n = 1$ and equals $\{\frac{1}{2}\}$ for $n = 2$.)
   - $\subset$: if x is in the union, then $\frac{1}{n} \le x \le 1 - \frac{1}{n}$ for **some** n, so $0 < x < 1$.
   - $\supset$: if $0 < x < 1$, choose n with $\frac{1}{n} \le \min\{x, 1 - x\}$. Then $x \in \left[\frac{1}{n}, 1 - \frac{1}{n}\right]$ for that n.
   - The endpoints 0 and 1 are in none of the intervals.
2. $\displaystyle\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$.
   - 0 is in every interval.
   - If $x \neq 0$, choose n with $\frac{1}{n} \le |x|$. Then $x \notin \left(-\frac{1}{n}, \frac{1}{n}\right)$, so x is not in **all** of them.
3. $\displaystyle\bigcap_{n=1}^{\infty} \left[0, 1 + \frac{1}{n}\right) = [0, 1]$.
   - If $0 \le x \le 1$, then $x < 1 + \frac{1}{n}$ for every n.
   - If $x > 1$, choose n with $\frac{1}{n} \le x - 1$. Then $x \notin \left[0, 1 + \frac{1}{n}\right)$.

- If $x < 0$, x is in none of the intervals.
- Note that the intervals are half-open but the intersection is closed: the point 1 is in every interval.

(b) Fix an outcome x. Then $x \in \bigcup_{n \geq k} A_n$ means "x is in some A_n with $n \geq k$," i.e., x is in at least one set **at or after** stage k. The intersection over k requires this for **every** k: no matter how far out we go, x belongs to some later A_n. That happens exactly when x belongs to **infinitely many** of the A_n.

So $\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n$ is the event "A_n **occurs infinitely often**," also called $\limsup_n A_n$. (If x were in only finitely many A_n, with the last one at $n = N$, then $x \notin \bigcup_{n \geq N+1} A_n$, so x fails at $k = N + 1$.)